BotaniX AI: Flower Species Classification using Deep Learning

Transfer learning with MobileNetV2 on the Oxford 102 Flowers dataset

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Problem and objective

- Many flower species look alike, so naming one from a photo is hard
- Task: classify a single photo into one of 102 species
- Goal: an accurate model built with transfer learning
- Goal: a web app where anyone can upload a photo and get the species
- Goal: code and weights shared in a public GitHub repository

Dataset: Oxford 102 Flowers

8,189
images in total

102
flower species

70 / 15 / 15
train / validation / test split

- Secondary dataset, loaded directly with torchvision
- 40 to 258 images per species, with varied light and backgrounds
- Official splits merged and re-split with seed 42
- Train 5,732 | Validation 1,228 | Test 1,229

Model and training

- MobileNetV2, pretrained on ImageNet
- Final layer replaced with 102 outputs; all layers fine-tuned
- Augmentation: random crop, flip, rotation, colour jitter
- About 2.4 million parameters; 9.4 MB as ONNX
- Best epoch chosen by validation accuracy

Setting	Value
Optimiser	**AdamW**
Learning rate	**3e-4, cosine schedule**
Batch size	**32**
Epochs	**10**
Loss	**Cross-entropy**
Hardware	**Colab T4 GPU**
Training time	**about 8.4 min**

Results

97.72%
test accuracy

0.9725
macro F1-score

28 / 1,229
test images misclassified

90.2%
validation accuracy after epoch 1

- Validation accuracy settles near 97.6% from epoch 7
- Training accuracy reaches 99.9%: mild overfitting, test result still strong
- Most errors are look-alike species, e.g. sword lily vs canna lily, petunia vs tree mallow

Web app and deployment

- Trained model exported to ONNX (9.4 MB, one file)
- Runs in the browser with ONNX Runtime Web: no server needed
- Same preprocessing as training: resize, centre crop, normalise
- Shows the top 3 species with confidence; warns when confidence is low
- Photo never leaves the user's device; hosted on Vercel
- Example: an oxeye daisy photo was predicted correctly at 96.4%

Conclusion and links

- 97.72% test accuracy after about 8.4 minutes of training
- A small pretrained model was enough for 102 species
- Future work: compare EfficientNet and ResNet, add Grad-CAM, add a not-a-flower class

Project links

Live app: https://botanix-flower-ai.vercel.app/

GitHub: https://github.com/MuhammadFaizan579/botanix-ai

Report:

https://botanix-flower-ai.vercel.app/Flower_Classification_Report.pdf